A mergeable measure of sentiment for text of any length

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Sentiment scores for text are usually bounded numbers, a class probability or a normalised word count. Such scores do not compose: the score of a chapter cannot be computed from the scores of its sentences, so long texts are cut and averaged, and the answer depends on the cut. Here I describe a measure for which granularity holds. Each word carries positive charge (hedons) and negative charge (dolors) taken from human word ratings, and a span of text is summarised by a signature of nineteen numbers that merges exactly with the signatures of its neighbours. One pass over a book gives the score of every chapter and sentence. Against human ratings in six corpora the calibrated score reaches a mean correlation of 0.51, level with the best lexicon method (0.49) and below fine-tuned transformers (0.59 and 0.68), at about 4,500 times their speed on a processor core. Two hypotheses from an earlier version were tested as pre-registered. Feeling is not front-loaded: what comes last counts most, a recency effect that three confirmatory tests support. Neutral words dilute a rating, but only when they follow it. Across 356 novels a conventional score moves from 0.11 to 0.76 with the size of the chunks, while merged signatures do not change. Essays written by language models differ from student essays on the same assignments mainly in an upbeat ending and in sparse use of intensifiers and negation. A model can be steered to an assigned profile of these measures, and doing so did not make its essays read more like the students'.

Sentiment analysis turns a piece of writing into a number that says how positive or negative it is. The number is computed for product reviews, posts, news, clinical notes and parliamentary speech, often for millions of texts at a time [1–5]. Two families of method are in use. Lexicon methods look each word up in a list of human word ratings and combine the values by rule [6–9]. Classifiers, now mostly fine-tuned transformers, are trained on labelled text and return a class probability [10–12].

Both families return a bounded number for each text: a probability, a mean, or a sum squashed into the interval from −1 to 1. A bounded score is intensive in the sense that temperature is intensive. It does not grow with the amount of text, and the value for a whole cannot be computed from the values of its parts. Three practical problems follow. A transformer reads a few hundred words at a time [13], so a long document is cut into chunks whose scores are averaged, and the average depends on where the cuts fall. A squashed sum saturates: in the novels studied below, VADER's compound score is beyond ±0.99 for 87% of whole chapters. And an average of averages can rank two bodies of text in the opposite order to their pooled values, the reversal known from Simpson's paradox [14–16].

Here I treat sentiment as a quantity that adds. Every word carries a positive and a negative charge taken from human word ratings. A span of text is summarised by a short vector, its signature, which holds counts and sums of charge weighted by powers of word position. Signatures of adjacent spans merge exactly into the signature of the joined span, so a novel is scored in one pass and every chapter, page and sentence can be read off afterwards without scoring anything again. Scores that people can compare, such as net charge per thousand words or a predicted human rating with an error bar, are read from the signature at whatever scale is wanted.

The parts have precedents. The hedonometer averages human word ratings [2,6]. Word-count tools and studies of mood keep positive and negative words apart [3,9,17], in line with evidence that the two are separable in evaluation [18]. Political scientists score texts from counts of opposed items on a log-ratio scale with a stated uncertainty [4,19]. Summaries that merge exactly across partitions, including sums weighted by polynomial or exponential functions of time, are standard in stream processing [20–23]. Emotional arcs of stories have been drawn with moving windows, transforms and fitted polynomials [24–26]. This paper joins these into one object for sentiment, states which position weights such an object can carry, and tests it against human ratings.

The aim is a mathematical measure of sentiment for which granularity holds: the score of a whole is the merge of the scores of its parts, at every level from the phrase to the book. The work began with a manuscript I wrote in 2024, which proposed that feeling in text could be scored as a number from word values, a decay over position and a penalty for words that carry no feeling. It made two claims about how people write and read. Front-loading: words that carry feeling come early in a text and count for more there. Dilution: words that carry none weaken the feeling of the whole. I wrote both down as hypotheses with decision rules, and committed them to the project repository, before scoring any evaluation data (Methods). Front-loading fails in every corpus examined. I replace it with a hypothesis of recency, that what comes last counts most, and three confirmatory tests on untouched data support it. Dilution holds, in a one-sided form that recency explains: neutral words weaken a rating when they follow the feeling and leave it almost unchanged when they come before it. A last experiment applies the measure to a current question and finds that essays written by language models differ from student essays on the same assignments in where and how they carry feeling.

Results

A signature that merges

Take a text as a sequence of words. Word i has a positive charge h_i and a negative charge d_i, each between 0 and 1, at most one of them nonzero. The values come from two open sets of human word ratings, the labMT list [2] (10,222 words, 50 raters each) and the VADER list [7] (7,494 entries, 10 raters each). On the 1,366 words they share, the two sets correlate at $r = 0.95$. I put both on one scale, on which 1 is the positive end of the rating scale and 0 its neutral point, and average them (Methods). Positive charge is measured in hedons (hd) and negative charge in dolors (dl), the names that moral philosophy uses for units of pleasure and pain [27,28]. One hedon is the charge of a word that every rater places at the top of the scale; *love* carries 0.85 hd and *terrible* 0.55 dl. A word rated within a quarter of the scale from neutral gets no charge and is called a buffer word, following the neutral band of the hedonometer [2]. In the six rated corpora used here, 23% of running words are charged.

The signature of a span holds the number of words L, the number

of charged words E, and the sums

$$S_k^+ = \sum_i h_i\, i^k, \qquad S_k^- = \sum_i d_i\, i^k, \qquad k = 0, \ldots, 3, \tag{1}$$

where i is the position of the word in the span, counted from zero. S_0^+ and S_0^- are the span's total hedons and dolors. The signature also holds the hedons and dolors that fall under a negation, after an intensifier, after a diminisher, and after a contrast word such as *but*, together with the sum of squared charges (Methods). That is 19 numbers, whatever the length of the span (Fig. 1a,b).

Signatures merge. If span A is followed by span B, the counts and totals add, and the position sums of B are shifted by the length of A:

$$S_k(AB) = S_k(A) + \sum_{j=0}^{k} \binom{k}{j} L_A^{k-j}\, S_j(B). \tag{2}$$

The operation is associative and can be undone, so the signature of B follows from those of AB and A (Proposition 1, Methods). A text of any length is therefore scored in one pass with constant memory. Parts can be scored separately and merged in reading order to the same result as a single pass. A table of prefix signatures, one row per sentence, gives the signature of any contiguous run of sentences from two of its rows. In integer arithmetic these statements hold to the last digit, and the reference implementation, which counts in thousandths of a hedon, is tested for exact equality on random texts. The fast implementation uses double-precision floats; over 356 novels its merged and single-pass signatures agree to a relative error below 1×10^{-13}.

Which position weights allow a merge? Suppose a summary keeps finitely many sums $\sum_i c_i\, \varphi(i)$ of charge against weight functions φ. The summary of AB can be computed from the summaries of A and B and their lengths exactly when the weight functions span a space that is closed under shifting the position, and such weights are combinations of $i^k \rho^i$: polynomials, exponentials and their products (Proposition 2, Methods). The statement is an instance of a classical fact about translation-invariant function spaces[29,30] and about functions on strings with finite-dimensional representations[31,32], and stream-processing systems rely on it[20]. For sentiment it says two things. The decay factor of the 2024 manuscript, a weight that falls geometrically with position, is one of the few position weights that survive a merge, at the cost of one number per decay rate. And among mergeable weights, polynomials alone can be re-expressed in relative position, that is, position divided by a length that is unknown until the text ends. The signature keeps polynomial sums for that reason: the sum of charge under any cubic function of relative position can be read from it, for a post and for a novel alike.

Reading a score from a signature

A signature is extensive. Scores that can be compared between texts are read from it (Fig. 1c).

Totals are the hedons $H = S_0^+$ and dolors $D = S_0^-$ over L words. *Density* is the net charge per thousand words, $1000\,(H - D)/L$; the hedonometer's average happiness is a density taken over charged words only.

Level follows a model from the psychology of judgement. In Anderson's averaging model a judgement is a weighted mean of the pieces of information received and an initial neutral impression[33], and information that says nothing pulls a judgement toward neutral, the dilution effect[34]. For words this gives

$$\text{level} = 1000\, \frac{H - D}{a + E + \kappa B}, \tag{3}$$

where $B = L - E$ counts buffer words, a is the weight of the neutral first impression in units of charged words, and κ is the weight of a buffer word relative to a charged one. With $\kappa = 0$ buffer words do nothing; with $a = 0$ and $\kappa = 1$ the level is the density. The denominator is the exposure, the amount of evidence behind the score, and it gives the level a standard error that shrinks as a text grows (Methods).

Calibrated valence is a predicted human rating. It is a linear combination of ten signature entries, each divided by the exposure: hedons, dolors, three position moments of net charge, and the charge under each operator. Given a and κ, the coefficients solve a ridge regression against human ratings, a convex problem with a closed-form solution; a and κ are chosen on a grid (Methods). The fitted coefficients say how much a negation flips a word and how position is weighed; they are estimates with intervals, in the tradition of rule constants measured from ratings[7,35].

The *arc* is the least-squares cubic fit of net charge against relative position. It has a closed form in the position sums (Methods), so it costs nothing beyond the signature.

Agreement with human ratings

I compared scores with human ratings on six corpora: sentences from six genres (EmoBank[36]), tweets (SemEval-2018[37]), Facebook posts[38], film-review sentences (the Stanford Sentiment Treebank, SST[39]), whole film reviews with the critic's rating[40], and Yelp reviews with star ratings[41]. The first five carry continuous ratings and form the primary set. Every constant was fitted leave-one-corpus-out: for each target corpus, a, κ and the ten coefficients came from the other five, and scores were then computed on the target's held-out test portion (Methods). The six fits chose a of 2 or 4 and κ of 0.2 or 0.5.

Averaged over the five primary corpora, the calibrated score correlates with human ratings at $r = 0.51$, against 0.49 for VADER[7], 0.44 for TextBlob and 0.42 for the hedonometer[2] (Fig. 2a; all values with intervals in Extended Data Table 1). The paired difference from VADER excludes zero in favour of the signature in 3 of six corpora and in favour of VADER in 0. The level of equation (3), which has no fitted coefficients, reaches 0.48. This meets a criterion that an earlier version of the method failed (Methods), by a small margin.

Methods trained on labelled text do better. A bag-of-words ridge regression trained on each corpus's own training portion averages 0.61. DistilBERT fine-tuned on SST-2[12] averages 0.59 and a RoBERTa model fine-tuned on tweets[42] 0.68. The signature is not a replacement for these where accuracy on one text matters most.

Removing parts of the signature shows what earns its place (Fig. 2c). Without the operator channels the mean correlation falls by 0.034; without dilution ($\kappa = 0$) by 0.015; without the neutral prior ($a = 0$) by 0.010. Merging hedons and dolors into one channel costs 0.002 and dropping the position moments 0.000: neither matters for agreement with raters, though both are needed for the readouts below.

The fitted coefficients have plain readings (shipped model, fitted on all six corpora; bootstrap intervals). A dolor moves the predicted rating 1.63 [1.54, 1.73] times as far as a hedon. A negated positive word counts as -1.60 [-1.85, -1.37] times its charge, about what the same amount of negative charge would do, while a negated negative word counts as -0.01 [-0.23, +0.20]: *not good* is bad, and *not bad* is neutral. An intensifier adds +1.02 [+0.85, +1.19] times the charge of the word it modifies. The data do not settle the effect of diminishers (-0.33 [-0.91, +0.09]) or of contrast words (-0.06 [-0.14, +0.02]); in particular they give no support to the rule that the clause after *but* counts for more[7].

Word charges can also be fitted. With the exposure and the operator constants held fixed, the predicted rating is linear in the vector of

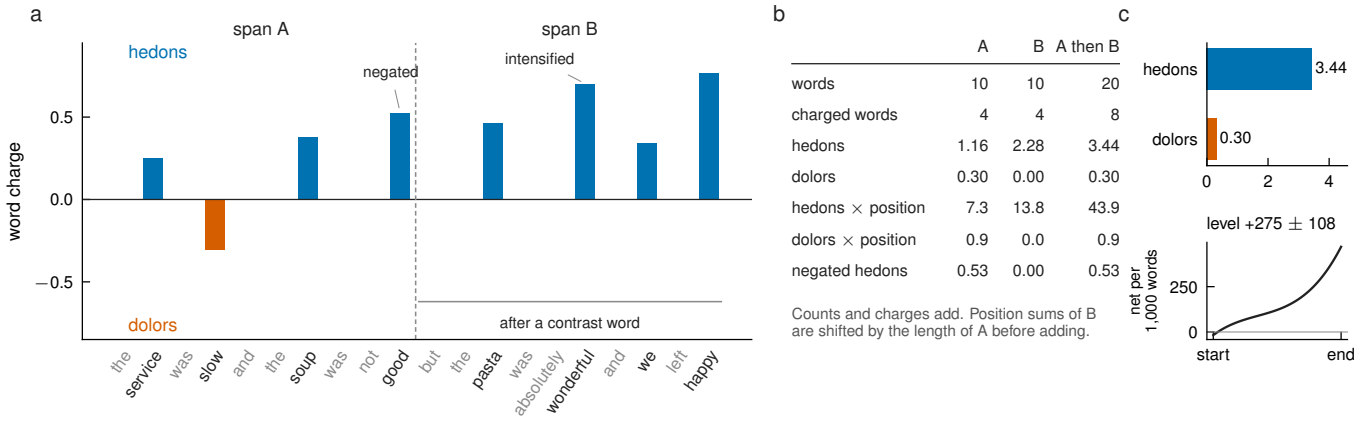


Figure 1 Charge, signature, merge, readouts. a, Two sentences scored word by word. Bars above the line are hedons, bars below are dolors; grey words are buffer words and carry no charge. *Good* sits inside the reach of *not*, *wonderful* follows an intensifier, and the second sentence follows a contrast word. **b**, Entries of the signature of each sentence and of the two read as one text, obtained by merging and equal to a single pass over both. **c**, Readouts of the merged signature: totals, the level with its standard error (shipped α and κ for the level), and the cubic arc of net charge over relative position.

charges, so the charges that best predict ratings, pulled toward the human word norms by a ridge penalty, solve one convex problem (Methods). Fitted leave-one-corpus-out, these refined charges raise the mean correlation to 0.55, the same as a bag-of-words ridge trained on the same five corpora (0.55). The gain is uneven. It is large on review text (SST 0.44 to 0.55, film reviews 0.29 to 0.49) and negative on the mixed genres of EmoBank (0.48 to 0.41) and on Facebook posts (0.62 to 0.58). Charges learned from ratings carry the genre of the texts they were learned from; human word norms travel better.

One hoped-for property did not hold. The standard error of a level shrinks with exposure, but the error of the calibrated score against human raters does not: in none of the six corpora did squared error fall as exposure rose (Extended Data Fig. 1a). The standard error says how well a text's own words pin down its level. It is not a prediction interval for a human rating.

Cost

Scoring is one table lookup per word. On one core of a desktop processor the implementation, written in Python with NumPy, scores 167,000 sentences per second, or 2.6 million words (Fig. 2b). On the same core VADER scores 27,900 sentences per second, TextBlob 11,100, DistilBERT 37 and the RoBERTa model 19. The signature is therefore about 4,500 times faster than DistilBERT, well past the factor of 100 that the plan set as its criterion. On a graphics card (RTX 3090, half precision) DistilBERT reaches 12,200 sentences per second, and one processor core running the signature is still 14 times faster. Time grows in proportion to the amount of text (log-log slope 1.02 from 10^4 to 10^6 sentences); ten million sentences took 59 seconds in fixed-size batches. Because signatures merge, the work divides among cores without changing any result, and twelve processes scored 0.8 million sentences per second. At these rates a billion sentences take 1.7 hours on one core or 21 minutes on twelve, against 23 hours for DistilBERT on the graphics card and 310 days on one core.

Where the feeling sits

Front-loading makes three predictions, and I tested each (Fig. 3).

The first is that charged words come early. They do not. The charge-weighted mean position, on a scale from −1 at the first word to +1 at the last, is above zero in all 8 kinds of text examined, from +0.002 in Yelp reviews to +0.105 in tweets, with every interval excluding zero (Extended Data Table 2). In EmoBank sentences, SST sentences and tweets the first tenth of the words carries 0.65 to 0.75 of the text's mean charge per word, and in those three and in Facebook posts the last tenth carries 1.34 to 1.55 (Fig. 3a). English sentences tend to open with determiners and pronouns and close on content. Facebook posts open at the mean (1.02). Yelp reviews are the one place with a charged opening, at 1.09 times the mean, matched by the close at 1.11. Negative charge sits later than positive charge in 7 of the 8.

The second prediction is that readers weigh early words more. The position kernel fitted in each corpus puts more weight on the first third of a text than on the last third in SST sentences (difference +0.26 [+0.14, +0.39]), less in tweets (-0.39 [-0.54, -0.24]) and Yelp reviews (-0.12 [-0.15, -0.09]), and is undecided in the other three (Fig. 3c). In the shipped model, fitted on all six corpora, the last third outweighs the first (-0.13 [-0.20, -0.07]).

The third test uses human ratings alone. The SST gives a human rating to every phrase of its parse trees [39], so the rating of a phrase can be regressed on the ratings of its two parts. Across 154,457 phrase pairs the later part gets the larger weight: 0.68 against 0.61 when both parts have at least two words (difference +0.073 [+0.055, +0.089]; Fig. 3b). The later part of an English phrase is often its predicate, so this is a fact about where evaluation sits in syntax as much as about order. Either way the early words do not dominate. Within EmoBank documents, rated intensity does not fall with sentence position either (slope +0.012 [-0.012, +0.032] per document length; 115 documents).

Front-loading is therefore refuted as a general rule. Earlier work on reviews found the same for whole documents, where the closing sentences carry the verdict [43–46], and it fits the recency effects known from memory and from retrospective judgements of experience [47,48].

Neutral words dilute, when they follow the feeling

Dilution fares better (Fig. 4). Holding a text's total charge fixed, rated intensity falls as the number of buffer words rises in 5 of six corpora, with partial rank correlations from -0.27 in EmoBank to -0.08 in SST sentences. Whole film reviews show no effect (+0.01 [-0.02, +0.04]). Held-out agreement peaks at a buffer weight κ between 0.2 and 1 in the same five corpora and at zero for film reviews (Fig. 4c).

Human phrase ratings show the effect without any scorer. Take a phrase with a clear rating and join it to a neutral phrase of m words. When the neutral words come after the rated phrase, the joined phrase keeps 0.78 of the rating for $m = 1$ and 0.62 [0.57, 0.67] for $m \geq 9$. When they come before it, the share kept is 0.75 for $m = 1$ and 0.76 [0.71, 0.80] for $m \geq 9$: no decline (Fig. 4a). The level of these shares is pulled down by regression to the mean, because the rated phrase

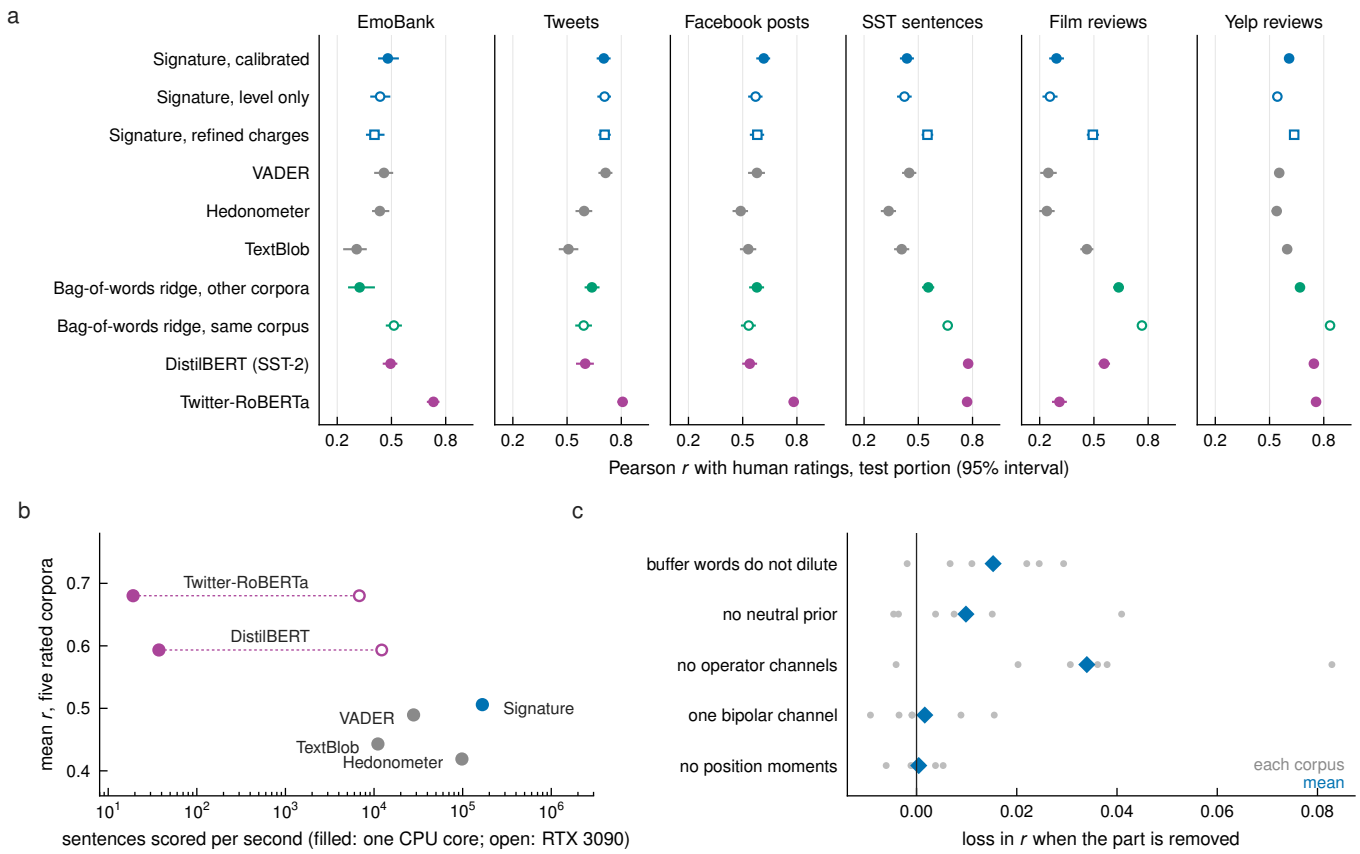


Figure 2 Agreement with human ratings, and cost. a, Pearson r between score and human rating on the held-out test portion of each corpus, with 95% bootstrap intervals. Blue: readouts of the signature, with constants fitted on the other five corpora (filled circle, calibrated valence; open circle, level; square, calibrated valence with refined word charges). Grey: lexicon methods. Green: bag-of-words ridge regression trained on the other five corpora (filled) or on the same corpus (open). Purple: transformer classifiers. **b**, Mean r over the five corpora with continuous human ratings against sentences scored per second on one CPU core. **c**, Loss in r when one part of the signature is removed and the constants are refitted.

was selected for a strong rating; the comparison between the two orders, and the trend with m, are not.

So the 2024 manuscript had dilution right and its direction of travel wrong. Trailing neutral words weaken a judgement and leading ones do not, which is recency once more: what a reader meets last counts most, and if that is neutral the feeling fades.

A revised hypothesis: recency

The two results point the same way, so I replace front-loading with a hypothesis of recency: what comes last in a text counts most toward a reader's judgement of it. The data above suggested it, which makes them useless as a test of it. I therefore wrote the hypothesis down with three predictions, committed them, and tested them on data that had played no part: the development and test trees of the treebank and 22,001 IMDB film reviews (Methods). All three held. In 11,474 held-out phrase pairs the later part has weight 0.69 and the earlier part 0.63 (difference +0.059 [+0.032, +0.085]). A rated phrase keeps 0.68 of its rating when two or more neutral words follow it and 0.77 when they come before it (difference +0.091 [+0.055, +0.127]). And the last fifth of an IMDB review separates favourable from unfavourable reviews better than the first fifth does (area under the curve 0.696 against 0.657; difference +0.039 [+0.031, +0.048]). Recency adds little to agreement with raters at the scale of a sentence, as the ablation of the position moments showed, but it holds from the phrase to the review.

A book at every scale

I scored 356 novels from Project Gutenberg (28.1 million words, median 25 chapters) sentence by sentence and merged the signatures up to chapters and books. Fig. 5a,b shows *Moby-Dick*: 213,523 words carrying 11,790 hd and 3,539 dl, a density of +39 per thousand words, with each of its 129 chapters and twelve spans of its final chapter read from the same pass.

Chapters differ by more than chance. The variance of chapter densities is 5.9 times (median; interquartile range 4.3 to 7.8) what results when a novel's sentences are shuffled across its chapters, and exceeds the shuffled value at $p \leq 0.01$ in 98% of novels (Fig. 5c). One number for a book hides most of what its chapters do.

The cubic arc read from a book's signature captures 41% (median) of the variance of an arc drawn with a moving window one tenth of the book wide; keeping moments up to degree nine would capture 63% (Fig. 5d). The two leading modes of variation among the window arcs of the 356 novels are close to the first two Legendre polynomials (cosine similarity 0.94 and 0.89), the rise or fall and the single turn that earlier studies of story shapes found[24]; the third mode is not close to the cubic (0.10).

Human labels exist for one body of narrative. Alm and colleagues labelled every sentence of 176 fairy tales with an emotion[49]. The cubic arc of each tale's merged signature agrees in sign with the cubic arc of its human labels in 70% of tales (median correlation 0.46; sign test $p = 6 \times 10^{-8}$; Extended Data Fig. 1c). For a tale's overall valence the picture is less favourable. The calibrated score of the merged

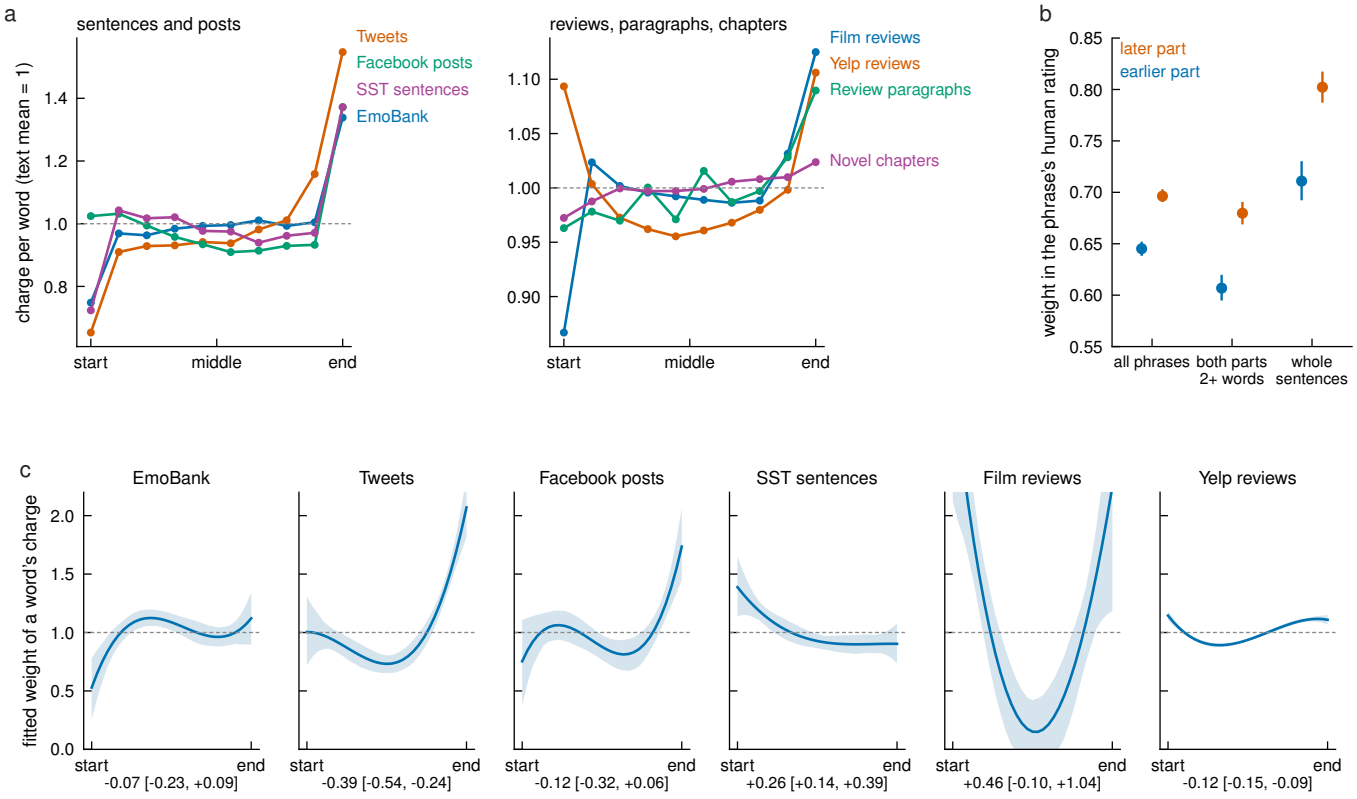


Figure 3 Feeling is not front-loaded. a, Charge per word in each tenth of a text, relative to the text's mean, for texts of ten or more words. **b**, Weights of the earlier and the later part in a regression of a phrase's human rating on the human ratings of its two parts, for binary nodes of the Stanford Sentiment Treebank (95% intervals, sentences resampled). **c**, Position kernel fitted against human ratings in each corpus: the weight given to a word's charge as a function of its relative position, with 95% bands. Below each panel, the mean weight over the first third minus the mean over the last third. The kernel for whole film reviews is poorly determined.

signature correlates with the mean human label at 0.50 [0.34, 0.63], and VADER averaged over sentences does better (0.63 [0.51, 0.73]). The human target here is itself a mean of sentence labels, which an average of sentence scores mirrors. VADER applied to the whole tale saturates beyond ±0.99 for 73% of tales and falls to 0.39.

A caution applies to every number in this section. Word norms rate words, not events. The final chapter of *Moby-Dick*, in which the ship is destroyed, scores near the book's own level, because *whale*, *sea* and *boat* are pleasant words to raters. Densities of narrative text are best read against the text's own baseline.

Pooling, averaging and chunking

Bounded scores depend on how a long text is cut (Fig. 6a–c). For 60 of the novels I computed a book score as the mean VADER compound score over chunks. The mean across books rises from 0.11 with one-sentence chunks to 0.76 with whole chapters, and 19% of pairs of books swap order between the two. Two transformer classifiers behave the same way on 300 whole film reviews: the score of the review and the mean score of its sentences differ in sign for 28% of reviews with DistilBERT and 51% with the RoBERTa model, and in both cases the mean of sentence scores predicts the critic's rating better than the score of the whole review does (r = 0.61 against 0.53, and 0.52 against 0.23); 97% of the reviews exceed the models' 512-token window and are truncated.

Averaging is the problem, whatever is averaged. If the densities of chunks are averaged, 8% of pairs of books swap order as the chunk size changes. If the chunks' signatures are merged, the book's density is the same for every chunking, to a relative difference below 1×10^{-14}.

How far apart are a pooled value and a mean of parts? For parts with exposures D_p and densities λ_p,

$$\frac{\sum_p D_p \lambda_p}{\sum_p D_p} = \bar{\lambda} + \frac{\mathrm{Cov}(D, \lambda)}{\bar{D}}, \tag{4}$$

an identity familiar from the decomposition of aggregate productivity[50]. The two differ when long parts differ in density from short ones. In novels they rarely do: the median gap between a novel's pooled density and the mean of its chapter densities is 4% of the standard deviation of density between novels, and 3.1% of the 63,190 pairs of novels are ordered differently by the two (Fig. 6d).

A difference between two bodies of text splits in the same spirit. Kitagawa's decomposition[51] writes a difference in pooled density as a part due to differences within strata plus a part due to the mix of strata. Four film critics in the review corpus[40] illustrate it, with the critic's own rating as the stratum (Fig. 6e). Pooled, Berardinelli's and Renshaw's reviews have the same density (difference -0.1 [-1.3, +1.1] per thousand words). Within the same rating Berardinelli's prose is the more negative by -1.2 [-2.5, +0.0], an interval that reaches zero, and his mix of ratings adds back +1.1 [+0.8, +1.6]. The decomposition is bookkeeping, and which part answers a question depends on the question[52]; the point is that a signature kept per stratum makes both parts available, and a pooled bounded score does not.

Human and machine writing

A measure that is cheap and does not depend on length can be pointed at a current question: does text written by a language model carry feeling differently from text that people write on the same prompt? Earlier comparisons mostly report that model output is less negative, and more neutral or positive, than human text[53–56], with exceptions by

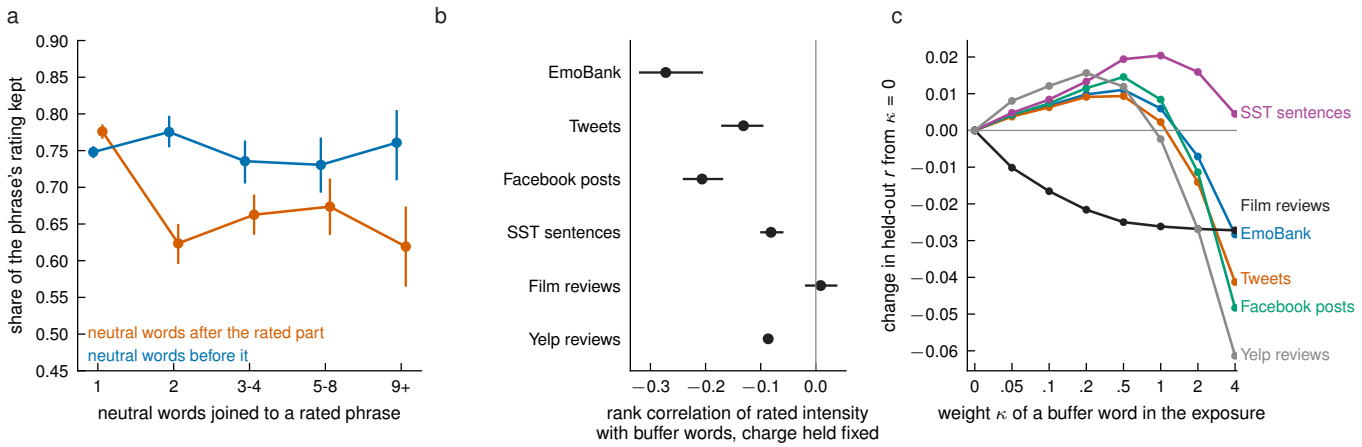


Figure 4 Neutral words dilute a rating when they follow it. a, Human phrase ratings from the Stanford Sentiment Treebank. A phrase with a clear rating is joined to a neutral phrase; the vertical axis is the share of the first phrase's rating that the joined phrase keeps, by the number of neutral words and their side (95% intervals). **b**, Partial rank correlation between rated intensity and the number of buffer words in a text, with total charge held fixed. **c**, Change in held-out r as the weight κ of a buffer word in the exposure moves away from zero, with coefficients refitted at each value.

domain[57,58], alongside differences in vocabulary and grammar[59,60]. Before generating or scoring any machine text I fixed ten length-free measures read from signatures and four predictions (Methods): machine text would be more positive overall (HL1), use fewer negative words (HL2), spread its charge more evenly over sentences (HL3), and close on a more positive note than its body (HL4).

Two comparisons were made. HC3 pairs questions with an answer from a person and an answer from ChatGPT[53]; 20,091 pairs had answers long enough to score. For essays, I sampled 160 essays that US school students wrote on eight assignments[61] and had four Claude models write to each student's assignment and length, once on the bare assignment and once with the instruction to write as a student of that grade would. The writing agents were told nothing about the study.

In HC3 all four predictions hold, and all four effects are small: ChatGPT answers are higher in net density by +0.14 human standard deviations, lower in dolor density by -0.06, lower in clustering by -0.04 and higher in ending lift by +0.13 (Fig. 7a). In the essays only the last holds under both instructions. The final fifth of a machine essay sits 16 per thousand words above the essay's own density, against 3 for the students, a difference of +0.60 [+0.24, +0.93] standard deviations (+0.52 [+0.22, +0.75] when the model imitates a student). Overall positivity goes against the prediction for plain essays, which are less positive than the students' (-0.23 [-0.40, -0.02]), and with it when the model imitates a student (+0.29 [+0.08, +0.52]). HL2 and HL3 are undecided for essays.

Two differences recur in all three comparisons, and I had predicted neither. Machine text puts less of its charge directly after an intensifier (-0.21, -0.58 and -0.48 standard deviations) and less of it under a negation (-0.26, -0.40 and -0.24, the last with an interval that reaches zero). People write *very* and *not* next to their feeling words more than these models do.

The differences are real and modest. A logistic regression on the ten measures tells human from machine text with an area under the curve of 0.63 in HC3, 0.74 [0.68, 0.82] for plain essays and 0.64 [0.56, 0.76] for essays written as a student (Fig. 7b). Word count alone does better in HC3 (0.72), where answer lengths differ, and is near chance for the length-matched essays (0.54). The regression fitted on HC3 scores 0.67 and 0.65 on the essays, so part of the pattern carries from ChatGPT answers of 2023 to Claude essays of 2026. None of this makes a detector. Purpose-built detectors use far richer evidence and still misfire, notably against non-native writers[62,63].

Can the measures steer a revision? For 57 plain machine essays, the model that wrote each draft revised it under four instructions: a control instruction, to make it read as a student of that grade wrote it; the control instruction plus a report naming each measure on which the draft lay outside the middle 80% of student essays on other assignments; and the control instruction plus one of two published prose skills written to remove the marks of machine writing (Methods). I score a text by its mean squared standardised distance from the centre of student essays (Fig. 7c). Drafts average 0.66 and control revisions 0.53. Revisions made with a skill average 0.54 and 0.49, neither distinguishable from the control (differences +0.01 [-0.07, +0.07] and -0.04 [-0.16, +0.04]): the skills work on wording and sentence shape, and these measures show no trace of them. Revisions that saw the report average 0.41. Their advantage over the control is -0.12 with an interval of [-0.25, +0.01] when assignments are resampled, as the plan specified, and [-0.20, -0.03] when essays are, so the planned test is undecided. The report moved what it named. It cut the mean ending lift from +0.31 to +0.02 standard deviations, against +0.21 for the control, and left intensifier use where it was (-0.60 to -0.57), because a draft low on intensifiers is still inside the wide human range and was not flagged.

A report of ranges is a weak instrument. It acts only on drafts outside a wide band, and if it worked perfectly it would pull every draft to one centre. A dial is stronger. Each of 60 further drafts was assigned the measured profile of one real student essay, drawn at random from other assignments, and was revised twice by its author model with a sheet giving the current and the target value of each measure; a control arm revised the same drafts twice with the plain instruction. The dial was followed. Across the 48 essays with both arms complete, achieved values correlate with assigned targets at a mean of 0.64 over the ten measures in the dial arm and 0.08 in the control arm (difference +0.56 [+0.39, +0.71]), from 0.83 for the best-followed measure to 0.24 for sentence clustering. The mean squared gap to the target fell from 1.67 to 1.07 after one round and 0.79 after two, while the control arm stayed at 1.55.

Following the dial did not make the essays more like student writing. The classifier's mean machine score is 0.64 for the drafts and 0.41 for the matched student essays; two plain revisions bring it to 0.43 and two rounds with the dial to 0.49 (difference +0.06 [-0.04, +0.17], undecided). A judge model that read both revisions of each draft

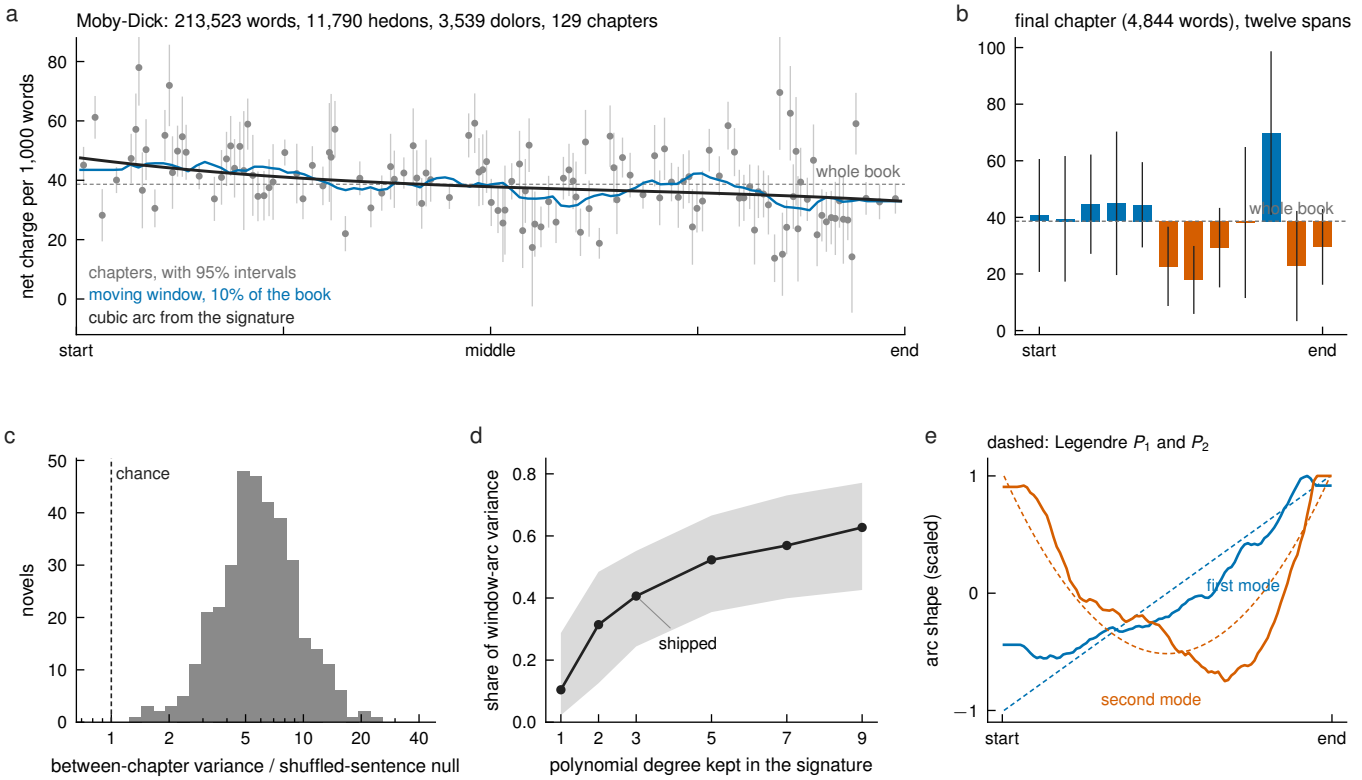


Figure 5 A novel at three scales from one pass. a, *Moby-Dick*: density of each chapter with 95% intervals from the word-level standard error, the cubic arc read from the book's signature, and a moving-window arc for comparison. **b**, The final chapter cut into twelve runs of sentences, each read from two rows of the prefix table; bars show the difference from the whole book's density. **c**, Between-chapter variance of density relative to its value when sentences are shuffled across chapters, for 356 novels. **d**, Share of the variance of the moving-window arc captured by the polynomial arc, by the highest degree kept (median and interquartile range). **e**, The two leading modes of variation among the novels' window arcs, with the first two Legendre polynomials.

without knowing which was which chose the dial revision as the better written in 62% of 48 pairs [43, 74] and as the one that reads more like a student's own work in 10% [5, 15]. Extended Data Table 4 shows the essay at the median improvement, chosen by rule and not by eye. To raise its positive charge the dial revision closes on *a fun, wonderful summer*, and its ending lift moved away from the target.

The measures therefore give a model a working control over how its text carries feeling, as far as a target can be reached by word choice in two passes. That is narrower than writing as a person does. A plain instruction to write as a student did as much for the classifier and far more for the judge.

One more observation belongs here. These experiments ask a model to make machine text resemble student writing, and the model sometimes refused: 3 of 240 requests in the first revision experiment, and 11 of 60 essays in the dial arm were left unfinished for that reason, 11 of them by the smallest of the four models. The refusals said that passing machine text off as a student's work misrepresents authorship. The tools released with this paper report tone against a human range for writers who disclose their use of a model, and the separations measured above are too weak for these measures to serve as a detector or to defeat one.

Discussion

The aim was a measure of sentiment for which granularity holds. The signature provides one: charge adds, position sums shift, and the signature of a whole is the merge of the signatures of its parts, exactly in integer arithmetic and to rounding error in floating point. Totals, density, a level with a standard error, a predicted rating and an arc are all read from the same nineteen numbers, and the scale at which to read them can be chosen after the text has been scanned.

No component of this is new. Averaged word norms[2], separate positive and negative counts[3,9], mergeable moment summaries[20,21] and the decompositions of pooled values[50,51] have long histories in their own fields, and the description of mergeable position weights is a classical result[29,30]. The paper adds their combination for sentiment, an implementation tested for exactness, and evidence on how position and neutral words enter human judgements.

Of the two hypotheses the work began with, one failed and one survived in altered form. Feeling is not front-loaded. The end of a phrase, a sentence or a review counts for more than its start, and three tests on untouched data supported that revised claim. Neutral words do dilute, as the averaging account of impression formation predicts[33,34], but mainly when they trail the feeling. A weighted average with a neutral first impression and weights that rise toward the end would describe all of these results. The shipped readout has the neutral prior and the buffer weight, and its position terms lean the same way.

The signature is not the most accurate reader of a single text. Transformers agree with raters better (0.59 to 0.68 against 0.51), and where the judgement of one text matters they are the better tool. The case for the signature rests on other properties: about 4,500 times the throughput per processor core, a result that does not change with the way a text is cut, and readouts that can be audited word by word. One sensible division of labour is to scan everything with signatures and to send a larger model only the spans that the scan marks.

Several limits qualify these results. Word norms rate words, not events, so densities of narrative text should be read against the text's own baseline, and irony, sarcasm and meaning carried by syntax beyond the four operators are missed. The lexicon is English and its

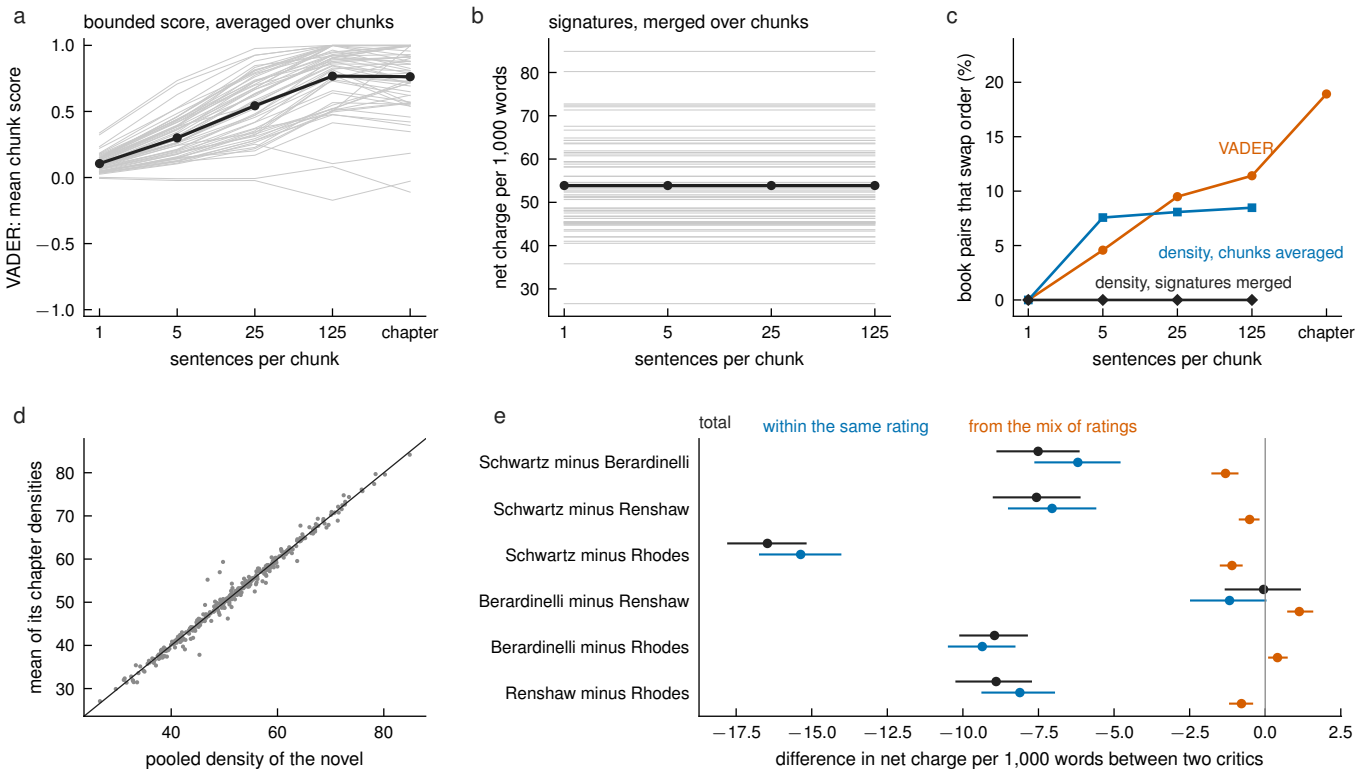


Figure 6 Averaging depends on the cut; merging does not. **a**, Book score for 60 novels computed as the mean VADER compound score over chunks of a given number of sentences (grey, each novel; black, mean). **b**, Density of the same novels from signatures of the same chunks, merged. **c**, Share of pairs of novels whose order changes relative to one-sentence chunks. **d**, Pooled density of each of 356 novels against the mean of its chapter densities (per 1,000 words). **e**, Differences in density between four film critics, split into the part within the same rating and the part due to the mix of ratings (95% intervals).

ratings date from the early 2010s. The constants were fitted on six corpora in which reviews and social media dominate, and charges refined on ratings did not travel across genres. No human ratings of chapters or books were available; above the sentence, validity rests on reviews and on fairy tales, where averaged VADER scores tracked overall valence better than the signature did. The standard error of a level is not a prediction interval for a human rating. Human judgements of long experiences are known to weigh peaks and endings rather than sums[48]; the signature keeps what an averaging or an end-weighted readout needs, and a readout of peaks would need signatures kept per part. The comparison of human and machine writing covers 160 student essays on eight assignments, four models from one family reached through an agent harness, and ChatGPT answers from 2023; its judge was a model, not a human reader. Analyses added after the first results are marked exploratory, and the confirmatory tests of recency, though committed in advance, come from the same kinds of text as the tests that suggested it.

Three additions would test the measure further: human ratings of chapters, so that readouts above the sentence can be validated directly; word norms in other languages; and the decayed sum that Proposition 2 allows, for a running level in streams of speech or messages.

Methods

Word charges. labMT 1.0 gives the mean of 50 ratings of happiness on a scale from 1 to 9 for 10,222 frequent English words[2]. The charge of a word is $c = (h - 5)/4$, so that $c = 1$ at the top of the scale, 0 at its neutral point and -1 at the bottom. The VADER list gives the mean of 10 ratings from -4 to 4 for 7,494 words, emoticons and abbreviations[7]. I put it on the labMT scale with one slope through the origin, the ratio of root-mean-square values over the 1,366 shared words, and averaged the two values where both exist. The result has 16,350 entries, of which 8,408 have $|c| \geq 0.25$ and carry charge; the rest are buffer words, as are words outside the list. Positive charge is $h = \max(c, 0)$ hedons and negative charge $d = \max(-c, 0)$ dolors. Charges are stored as integers in thousandths. Adding the NRC-VAD[64] and Warriner[65] norms ($r = 0.88$ and 0.92 with labMT on shared words) raised mean agreement on development data from 0.521 to 0.534; their licences do not allow redistribution, so they are not part of the shipped lexicon and no reported result uses them.

Tokens and operators. Text is lower-cased. Emoji are replaced by their names, using the list distributed with VADER, and 40 emoticons are recognised when they follow a space. Words are runs of letters and apostrophes. A full stop, exclamation mark or question mark ends a sentence; a comma, semicolon or colon ends a clause. Four closed lists of operator words carry no charge themselves: negators (*not*, *never*, forms ending in *n't* and others), intensifiers (*very*, *really*, *so*), diminishers (*slightly*, *somewhat*) and contrast words (*but*, *however*, *yet*, *nevertheless*, *nonetheless*). A negator reaches the next three words or the next punctuation mark, whichever comes first; an intensifier or diminisher reaches the next two words; a contrast word reaches the end of its sentence. Every sentence end clears all operators. The scan is a finite-state transducer and uses string replacement, translation and splitting only.

Signature. For a span of L words with charges h_i, d_i at positions $i = 0, \ldots, L-1$, the signature holds L; the number E of charged words; $S_k^{\pm}$ for $k = 0, \ldots, 3$ (equation 1); the hedons and the dolors inside the reach of each of the four operators; and $\sum_i (h_i - d_i)^2$. The fast implementation computes signatures for a batch of texts with array operations. A reference implementation loops over tokens in integer arithmetic and defines the result; the test suite checks that the two agree, and checks the propositions below for exact equality on random texts.

Proposition 1. Let A and B be texts and let A end at a sentence end. Then $\sigma(AB) = \sigma(A) \oplus \sigma(B)$, where $\oplus$ adds every entry except the position sums and applies equation (2) to those. The operation $\oplus$ is associative, the zero vector is its identity, and for fixed $\sigma(A)$ the map $\sigma(B) \mapsto \sigma(A) \oplus \sigma(B)$ is invertible.

Proof. Operators do not reach across a sentence end, so each word of B has the same charge and the same operator context in AB as in B. Counts and totals therefore add. A word at position j of B is at position $L_A + j$ of AB, and expanding $(L_A + j)^k$ gives equation (2). Write T_ℓ for the linear map that equation (2) applies to the position sums for a shift by ℓ. It is the matrix of the substitution $x \mapsto x + \ell$ on polynomials of degree below four in the monomial basis, so $T_\ell T_m = T_{\ell+m}$ and $T_\ell^{-1} = T_{-\ell}$, which give associativity and invertibility. □

Merging is not commutative: parts can be scored in any order and on any machine, and must be merged in reading order. With prefix signatures $\sigma_j = \sigma(s_1 \cdots s_j)$ over sentences, the signature of sentences i to j is $T_{-L}(\sigma_j - \sigma_{i-1})$ in the position sums, with

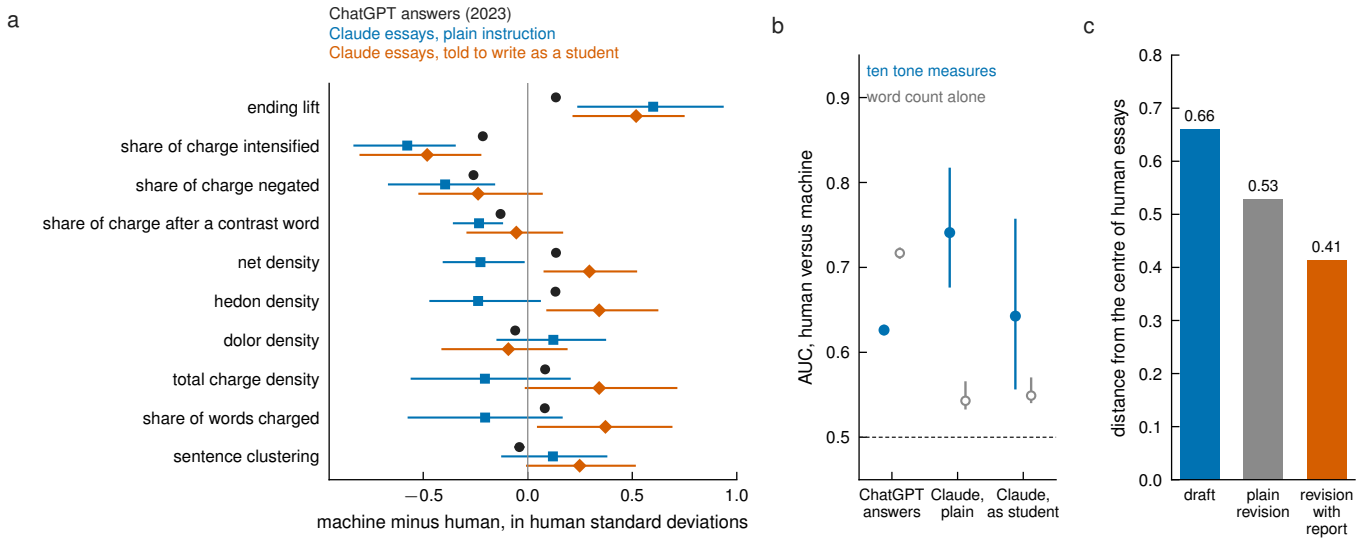


Figure 7 Human and machine writing on the same prompts. a, Difference between machine text and human text on ten length-free measures, in human standard deviations, with 95% intervals: 20,091 questions answered by people and by ChatGPT, and 160 student essays matched by essays from four Claude models under two instructions. **b**, Area under the curve of a logistic regression that tells human from machine text using the ten measures (blue) or word count alone (grey), scored out of fold. **c**, Mean squared standardised distance from the centre of student essays for machine drafts and for revisions of the same drafts.

L the length of the first $i-1$ sentences. In floating point this difference loses precision for high moments of short spans deep inside long texts; the integer implementation does not.

Proposition 2. Let $\varphi_1, \dots, \varphi_n$ be linearly independent real sequences on positions $0, 1, 2, \dots$, and for a real charge sequence c of length L let $\Phi(c) = (L, \sum_i c_i \varphi_1(i), \dots, \sum_i c_i \varphi_n(i))$. There is a function F with $\Phi(cc') = F(\Phi(c), \Phi(c'))$ for all charge sequences c, c' if and only if the span V of the φ_r is closed under the shift $(T\varphi)(i) = \varphi(i+1)$. In that case every φ_r satisfies a linear recurrence with constant coefficients of order at most n, and is a combination of sequences $i^k \rho^i$, with ρ a nonzero complex number, and of sequences that vanish from position n onward.

Proof. If V is closed under the shift, then $T^\ell \varphi_r = \sum_s M_{rs}^{(\ell)} \varphi_s$ for some matrices $M^{(\ell)}$, and $\sum_j c'_j \varphi_r(L+j) = \sum_s M_{rs}^{(L)} \sum_j c'_j \varphi_s(j)$, so $\Phi(cc')$ follows from $\Phi(c)$ and $\Phi(c')$. Conversely, fix $\ell \geq 1$ and let c be ℓ zeros. Take J large enough that the restrictions of the φ_s to positions below J are linearly independent, and let c' range over sequences of length J. By assumption $\sum_j c'_j \varphi_r(\ell + j)$ depends on c' only through the numbers $\sum_j c'_j \varphi_s(j)$. Adding to c' any d with $\sum_j d_j \varphi_s(j) = 0$ for all s therefore leaves it unchanged, so the linear functional $d \mapsto \sum_j d_j \varphi_r(\ell + j)$ vanishes on the common kernel of the n functionals $d \mapsto \sum_j d_j \varphi_s(j)$ and is a linear combination of them. The coefficients are unique, and do not change with J because the equations for J are among those for $J+1$. Hence $T^\ell \varphi_r \in V$. The shift is then a linear map on the n-dimensional space V and, by the Cayley–Hamilton theorem, is annihilated by its characteristic polynomial p, so every member of V satisfies the recurrence $p(T)\varphi = 0$. Its solutions are spanned by $i^k \rho^i$ for the nonzero roots ρ of p and, for a zero root of multiplicity m, by the sequences supported on positions below m. □

The proposition concerns summaries that are linear in the charges. It is the sequence form of the description of finite-dimensional translation-invariant function spaces [29,30], and of the fact that a function on strings has a finite-dimensional linear representation exactly when its Hankel matrix has finite rank [31,32]. A geometric weight ρ^i gives the one-number case $S_\rho(AB) = S_\rho(A) + \rho^{L_A} S_\rho(B)$. On relative position: a weight $g\big((i+\tfrac{1}{2})/L\big)$ is available from a summary for every length L only if the span of weights is closed under rescaling of position as well as under shifts. A finite-dimensional space of functions of a real variable with both properties is the space of polynomials up to some degree: closure under shifts gives exponential polynomials [29], rescaling rules out any nonzero exponent λ because the functions $e^{\lambda s x}$, $s > 0$, are linearly independent, and a shift-invariant space of polynomials contains the derivatives of its members.

Readouts. Density is $1000\,(H - D)/L$. Level is equation (3); its standard error is $1000\sqrt{\sum_i (h_i - d_i)^2 - (H-D)^2/L}\,/\,(a + E + \kappa B)$, which treats the charges of a text's words as independent draws. With $x_i = (2i+1)/L - 1$ and Legendre polynomials P_k, the moments $m_k = \sum_i (h_i - d_i) P_k(x_i)$, $k \leq 3$, are linear combinations of the position sums, and the arc is $f(x) = \sum_k \frac{2k+1}{L} m_k P_k(x)$. The sampled polynomials are orthogonal up to terms of order L^{-2}, so f is the least-squares cubic to that order; over the novels it agrees with a direct fit to 10^{-8}. The calibrated valence is $b_0 + \mathbf{b} \cdot \mathbf{x}/(a + E + \kappa B)$, where $\mathbf{x}$ holds H, D, the moments m_1, m_2, m_3, the negated hedons, the negated dolors, and the net charge that is intensified, diminished, and after a contrast word.

Corpora. Table 1 lists the rated corpora. Official training, development and test portions were kept where they exist. Facebook posts and film reviews were split 40/10/50 by a seeded shuffle. Yelp training and development portions are seeded samples of the official training set.

Table 1 Rated corpora. Numbers of texts in the training, development and test portions.

Corpus	Unit	Human rating	Train / dev / test
EmoBank [36]	sentence, six genres	valence, 1 to 5	8,062 / 1,000 / 1,000
SemEval-2018 [37]	tweet	valence, 0 to 1	1,181 / 449 / 937
Facebook posts [38]	post	valence, 1 to 9	1,158 / 289 / 1,448
SST [39]	sentence	sentiment, 0 to 1	8,544 / 1,101 / 2,210
Film reviews [40]	review	critic's rating	2,002 / 501 / 2,503
Yelp [41]	review	stars, 1 to 5	100,000 / 10,000 / 50,000

Fitting and evaluation. Ratings were standardised within each corpus with the mean and standard deviation of its training portion, and corpora were given equal total weight. For each target corpus and each pair (a, κ) on the grid $a \in \{0, 0.5, 1, 2, 4, 8, 16, 32\}$, $\kappa \in \{0, 0.05, 0.1, 0.2, 0.5, 1, 2\}$, the coefficients were fitted by ridge regression on the training portions of the other five corpora (penalty 10^{-3} times the mean diagonal of the weighted Gram matrix; intercept unpenalised). The pair with the highest mean Pearson r on those corpora's development portions was kept, the coefficients were refitted on training plus development portions, and the target's test portion was scored once. The shipped constants ($a = 4$, $\kappa = 0.5$) are the same fit on all six corpora. Intervals are 95% percentile intervals from 2,000 bootstrap resamples, drawing source documents for EmoBank and items elsewhere; differences between methods use the same resamples. Comparators: VADER compound score; the hedonometer, the mean labMT value of words outside its neutral band [2]; TextBlob polarity; ridge regression on TF-IDF weights of words and word pairs, trained on the target's own training portion or on the other five corpora; and two transformer classifiers scored as $P(\text{positive}) - P(\text{negative})$ on text truncated to 512 tokens, DistilBERT fine-tuned on SST-2 [12] and a RoBERTa model fine-tuned on tweets [42]. The transformers were run on a seeded sample of 5,000 Yelp test reviews, and for Yelp all methods are compared on that sample.

Refined charges. With exposure $a + L$ and operator multipliers θ from a one-channel fit, the predicted rating of text t is $b_0 + s \sum_i \mu_i c_{w_i} / (a + L_t)$ with $\mu_i = 1 + \theta \cdot (\text{operator flags of word } i)$, which is linear in the vector c of word charges. The refined charges minimise the weighted squared error plus $\lambda \| s(c - c^0) \|^2$, where c^0 holds the norm charges and words that occur in at least 25 training texts start at zero. The penalty $\lambda \in \{0.03, 0.1, 0.3\}$ was chosen on development data of the fitting corpora, and the readout was then refitted with the refined charges. Each step is a ridge regression; the sequence is not a joint optimum.

Position and dilution. For where charge sits, each text of at least ten words contributes $\sum_i |c_i| x_i$ and $\sum_i |c_i|$, and the statistic is the ratio of their sums over texts. Besides the six rated corpora, the units were paragraphs of the film reviews and chapters of 150 of the novels. The position kernel of a corpus is $1 + (b_1 P_1 + b_2 P_2 + b_3 P_3)/b_0$ from a one-channel fit to all its portions with the shipped a and κ; the contrast is its mean over the first third minus its mean over the last third, with intervals from 1,000 resamples. For the treebank, every binary node of the training trees whose phrase and both parts have ratings in the phrase dictionary was used (100% of nodes); ratings were centred at 0.5, the phrase rating was regressed on the two part ratings, and sentences were resampled. For dilution in the treebank, a part counts as rated if its centred rating is at least 0.15

in absolute value and as neutral if at most 0.05; the share kept is the ratio of sums of the signed phrase rating to the absolute part rating. For dilution in the corpora, rated intensity is the absolute deviation of a rating from the corpus median, and the partial rank correlation with the number of buffer words removes the rank of total charge from both by linear regression. For EmoBank documents of eight or more sentences, $|V - 3|$ was regressed on relative sentence position with document means removed, resampling documents.

Novels, fairy tales and critics. From a seeded sample of English fiction in the Project Gutenberg catalogue, 640 texts were fetched from a mirror, and 356 had at least ten chapters of 300 or more words and 20,000 words in all after removal of front matter. Chapters were found by headings and closed with a full stop. The null for chapter differences shuffles a novel's sentences 200 times and keeps the number of sentences in each chapter. Arcs were compared on 100 points; the moving window is 10% of the words. The fairy-tale annotations [49] give two annotators' emotion labels for each sentence; a label of happy or positively surprised was scored +1, neutral 0, and any other emotion −1, and the two were averaged. Critic comparisons use all 5,006 reviews with the four-level rating as stratum; intervals resample reviews within critic.

Human and machine writing. The ten measures are: hedons, dolors, net charge and total charge per thousand words; the percentage of words that are charged; the percentage of charge that is negated, intensified, and after a contrast word; ending lift, the mean of the cubic arc over the last fifth of the text minus the text's density; and sentence clustering, the between-sentence variance of density divided by its expected value if the text's words were dealt to its sentences at random, $\sigma^2(m-1)/L$ for m sentences, L words and word-level variance σ^2 of net charge. From HC3 [53] I took the first human answer and the first ChatGPT answer to each question and kept the 20,091 questions where both have at least 30 words; contractions split by the corpus's tokenisation of human answers were rejoined. From PERSUADE 2.0 [61] I drew 20 essays of 150 to 800 words for each of the eight assignments that need no source text (160 essays, seed 0). Each essay was matched by two machine essays written to the same assignment and word count, one with no further instruction and one with the instruction to write as a student of the same grade would in a timed test. The writers were subagents of the coding assistant, one essay per agent, running Claude Fable 5.2, Opus 5.5, Sonnet 5.5 and Haiku 5.5 in rotation at low reasoning effort. Each agent received only its task; none was told of the study or its hypotheses. Two of the 320 requests produced no essay. Differences are paired by essay or question and standardised by the human standard deviation; intervals resample assignments (eight) for the essays, as planned, and questions for HC3; results that resample essays are given where they differ. The classifier is a logistic regression on the ten standardised measures, scored out of fold with all texts of an assignment or question in one fold. For the tone check, 60 plain-instruction drafts (15 per model) were each revised by the model that wrote them under four instructions: the control instruction, to revise the draft so that it reads as a student of that grade wrote it; the control instruction plus a report listing each measure on which the draft lay outside the middle 80% of student essays on other assignments; and the control instruction plus an instruction to do the rewrite with one of two published prose skills for removing the marks of machine writing. Distance is the mean over measures of the squared standardised difference from the mean of student essays on other assignments. The model declined 3 of the 240 revision requests, stating that making machine text read as a student's work would misrepresent authorship; essays with all four revisions (57) enter the comparison.

Throughput. Timings were taken on a rented machine (AMD Ryzen 5 5600X 6-Core Processor, twelve threads; RTX 3090) so that processor and graphics-card figures come from one host. The plan had named the author's laptop, which was replaced because long runs overheated it. The texts are 200,000 sentences from Yelp reviews (mean 16 words). Each method scored a prefix of the same list: all of it for the signature, 20,000 sentences for the hedonometer, 5,000 for VADER and TextBlob, 600 for each transformer on one thread, 3,000 on twelve threads and 20,000 on the graphics card (batches of 128 sorted by length, single and half precision). Processor timings are the best of repeated runs.

Pre-specification. The method, the corpora, the outcomes, the comparators and the decision rules for the hypotheses were written into an analysis plan and committed to the repository (commit `7524c58`) before any evaluation data were scored. Test portions were first scored with the plan as written. Deviations since then are listed in the plan: the grid for α and κ was widened after the first run chose its upper edge, without changing the choice; the tokeniser was rewritten for speed; a bug in the selection of constants for the uncalibrated level was fixed; refined charges, the bag-of-words ridge trained on other corpora, the treebank analyses and the check of error against exposure were added and are exploratory. The comparison of human and machine writing was specified in an addendum committed before any machine text was generated (commit `4b10b12`). A February 2026 version of the method, built on word embeddings, failed its own criterion of matching VADER on GoEmotions; the criterion carried over here is a mean r over the primary corpora at least that of VADER.

Software. Python 3.10, NumPy 2.0, SciPy 1.15, scikit-learn 1.5, PyTorch 2.11 and Transformers 4.51. Figures were drawn with Matplotlib 3.9 and SciencePlots [66].

Use of AI tools. The aims and the two hypotheses come from the author's 2024 manuscript. The code, the analyses and the text of this draft were produced with Claude (Anthropic), an AI assistant working in a coding environment under the author's direction, and were reviewed by AI reviewer agents whose reports are in the repository. The essays in the comparison of human and machine writing were written by Claude models by design. Responsibility for the content rests with the author.

Data availability

All corpora and lexicons are public and are downloaded by `scripts/fetch_data.py`; none is redistributed except the open lexicon derived from labMT (CC BY) and the VADER word list (MIT). The machine-written essays, the result files behind every number and figure, and the analysis plan are in the repository.

Code availability

`https://github.com/ayazkhan27/anonymity`. Every number in the text is generated from the result files by `scripts/make_numbers.py`.

Declarations

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Extended data

Extended Data Table 1 Pearson r with human ratings on held-out test portions, with 95% bootstrap intervals. The mean is over the five corpora with continuous human ratings (all but Yelp). Signature constants are fitted on the other five corpora unless stated.

	EmoBank	Tweets	Facebook	SST	Film reviews	Yelp	Mean
n (test portion)	1,000	937	1,448	2,210	2,503	5,000	
Signature, calibrated	0.48 [0.43, 0.54]	0.70 [0.67, 0.73]	0.62 [0.58, 0.65]	0.44 [0.40, 0.47]	0.29 [0.26, 0.33]	0.61 [0.59, 0.62]	0.51
Signature, level only	0.44 [0.39, 0.49]	0.71 [0.67, 0.74]	0.57 [0.54, 0.60]	0.42 [0.39, 0.46]	0.26 [0.22, 0.29]	0.54 [0.52, 0.56]	0.48
Signature, constants from the same corpus	0.48 [0.43, 0.55]	0.73 [0.70, 0.76]	0.62 [0.58, 0.65]	0.46 [0.43, 0.49]	0.40 [0.37, 0.44]	0.63 [0.61, 0.64]	0.54
Signature, refined charges	0.41 [0.36, 0.46]	0.71 [0.68, 0.74]	0.58 [0.55, 0.61]	0.55 [0.52, 0.58]	0.49 [0.47, 0.52]	0.64 [0.62, 0.65]	0.55
VADER	0.46 [0.41, 0.50]	0.71 [0.68, 0.75]	0.58 [0.54, 0.62]	0.45 [0.42, 0.48]	0.25 [0.21, 0.29]	0.55 [0.54, 0.57]	0.49
Hedonometer	0.44 [0.40, 0.48]	0.59 [0.55, 0.63]	0.49 [0.45, 0.52]	0.34 [0.30, 0.37]	0.24 [0.20, 0.28]	0.54 [0.52, 0.56]	0.42
TextBlob	0.31 [0.24, 0.36]	0.51 [0.46, 0.56]	0.53 [0.49, 0.57]	0.41 [0.37, 0.44]	0.46 [0.43, 0.49]	0.60 [0.58, 0.61]	0.44
Bag-of-words ridge, other corpora	0.32 [0.27, 0.40]	0.64 [0.60, 0.67]	0.58 [0.54, 0.61]	0.56 [0.53, 0.58]	0.64 [0.61, 0.66]	0.67 [0.65, 0.68]	0.55
Bag-of-words ridge, same corpus	0.51 [0.47, 0.55]	0.59 [0.55, 0.63]	0.53 [0.50, 0.57]	0.66 [0.64, 0.68]	0.77 [0.75, 0.78]	0.83 [0.83, 0.84]	0.61
DistilBERT (SST-2)	0.50 [0.46, 0.53]	0.60 [0.56, 0.64]	0.54 [0.50, 0.57]	0.78 [0.76, 0.79]	0.56 [0.53, 0.58]	0.75 [0.73, 0.76]	0.59
Twitter-RoBERTa	0.73 [0.70, 0.76]	0.81 [0.78, 0.83]	0.78 [0.76, 0.80]	0.77 [0.75, 0.79]	0.31 [0.27, 0.34]	0.76 [0.75, 0.77]	0.68

Extended Data Table 2 Charge-weighted mean position of words, from -1 (first word) to $+1$ (last word), for texts of ten or more words, with 95% intervals; and the same for hedons and dolors separately.

Unit	Texts	All charge	Hedons	Dolors
EmoBank	5,975	+0.060 [+0.052, +0.068]	+0.051	+0.107
SemEval-2018 tweets	2,127	+0.105 [+0.096, +0.114]	+0.090	+0.156
Facebook posts	1,698	+0.021 [+0.012, +0.031]	+0.018	+0.036
SST sentences	9,358	+0.050 [+0.046, +0.055]	+0.042	+0.087
Film reviews	5,006	+0.023 [+0.022, +0.025]	+0.024	+0.022
Yelp reviews	157,452	+0.002 [+0.001, +0.002]	-0.001	+0.023
Film review paragraphs	22,514	+0.017 [+0.016, +0.019]	+0.011	+0.041
Novel chapters	2,583	+0.007 [+0.005, +0.009]	+0.005	+0.015

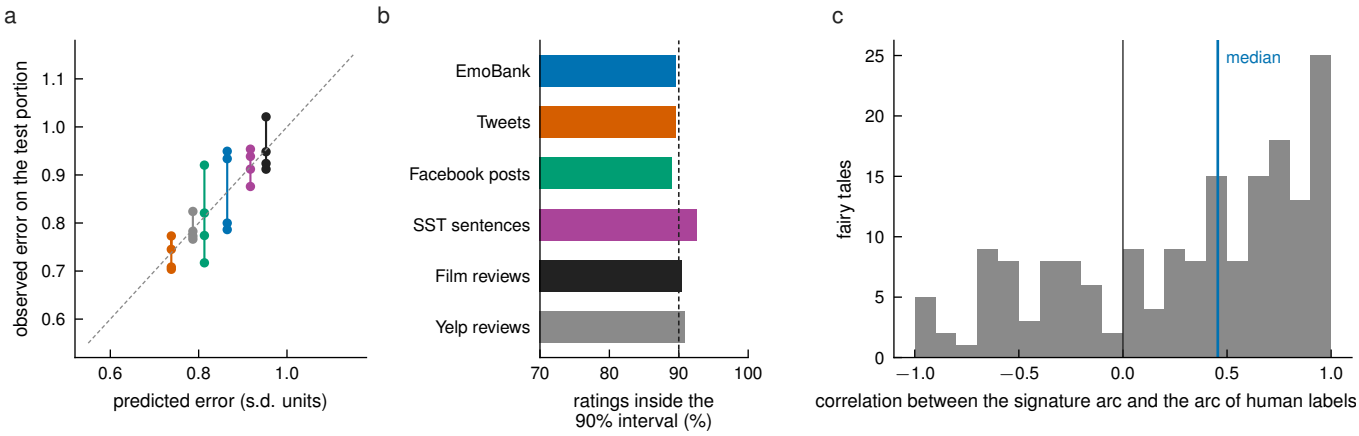
Extended Data Table 3 Coefficients of the shipped calibrated readout ($a = 4$, $\kappa = 0.5$), fitted on six corpora. The readout predicts a human valence rating in standard-deviation units.

Signature entry (per unit exposure)	Coefficient
intercept	-0.356
hedons	+5.850
dolors	-9.554
net charge $\times P_1$	+0.880
net charge $\times P_2$	+1.329
net charge $\times P_3$	+1.259
negated hedons	-15.196
negated dolors	+9.643
intensified net charge	+7.817
diminished net charge	-2.533
net charge after contrast	-0.470

Extended Data Table 4 The dial at work on one essay, chosen as the essay at the median improvement in gap to its target (model: Sonnet; assignment: Seeking multiple opinions). First and last 45 words of the machine draft, of its revision after two rounds with the dial, and of its revision after two plain rounds; and five of the measures with the assigned target.

Machine draft	After two rounds with the dial	After two plain revisions
Opening. When I had to choose between two summer programs last year, my first instinct was to ask my mom and go with whatever she said. She thought the robotics camp was a safe bet because it looked good for college. But then I talked to . . . *Closing.* . . . it. But asking a few trusted people, especially people with different experiences, gives you more information, exposes hidden bias, and sharpens your own thinking. That is why I would always rather ask several people than rely on one, even when the first answer sounds convincing.	*Opening.* When I had to choose between two summer programs last year, my first idea was to ask my mom and do whatever she said. She thought the robotics camp was a safe choice because it would look great for college. Then I talked to my . . . *Closing.* . . . the end the choice belongs to the person who has to live with it. I feel glad I did not ask just one person. Kids who ask a few trusted people get a fair chance to choose, and that makes for a fun, wonderful summer.	*Opening.* Last summer I had to choose between two summer programs. At first I just wanted to ask my mom and do whatever she said. She told me the robotics camp was the safe choice because it would look good on my college applications. But then . . . *Closing.* . . . you ask a few people you trust, especially people who have had different experiences, you get more information and you can spot bias. That is why I would always ask a few people instead of just one, even if the first answer sounds really good.

Measure	Draft	Target	Dial	Plain
Hedons per 1,000 words	90.1	105.6	113.5	94.3
Dolors per 1,000 words	4.8	15.1	6.3	4.5
Charge negated (%)	4.7	7.3	7.6	5.7
Charge intensified (%)	0.8	2.0	0.0	3.7
Ending lift (per 1,000 words)	8.2	-19.7	12.7	-0.2



Extended Data Fig. 1 Error against exposure, and arcs of fairy tales. a, Error of the calibrated score against human ratings on each test portion, in quartiles of exposure, against the error predicted by a variance model with a term that shrinks with exposure; the fitted model kept no such term, and observed error does not fall with exposure. **b**, Share of ratings inside the 90% interval implied by the fitted variance. **c**, Correlation between the cubic arc of a tale's merged signature and the cubic arc of its sentence labels from two human annotators, for 176 fairy tales.